Chapter-03
Presentation of Data

At the end of this chapter, students will be able to –

i. Explain why presenting data visually or tabularly is essential for quick comprehension, pattern recognition, and effective decision-making.

ii. Distinguish between tabular representation (such as frequency tables) and graphical or diagrammatic representation (such as charts and graphs).

iii. Organize raw, classified data into well-structured frequency tables using appropriate class intervals, boundaries, and tally marks.

iv. Choose the correct chart or graph type (e.g., bar charts for qualitative data, histograms or frequency polygons for quantitative continuous data) based on the nature of the variables.

v. Extract key trends, comparisons, proportions, and statistical summaries directly from various graphical formats.

vi. Identify common design errors, scale distortions, or formatting biases in data visualization that can mislead audiences.

For further statistical analysis, raw data are often classified according to the variables such that

1. Qualitative data,
2. Discrete data, and
3. Continuous data.

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Once classification has been done, the classified data can be condensed and summarized in two basic forms:

(i) Tabular form (known as Frequency distribution), and

(ii) Diagrammatical and graphical form.

Condensing and summarizing Data

From the statistical point of view, frequency distribution is the most important tabular form of condensing and summarizing a large mass raw statistical data. Some concepts related to frequency distribution are briefly described below. Illustration of the concepts is provided later with an example.

Frequency. Number of times an observation occurs in a set of data is called the frequency of that observation. Number of observations in a class is called the frequency of that class or category.

Frequency distribution. A frequency distribution is a tabular summary of data where observations are divided into different non overlapping classes or categories and frequency of each class or category is arranged accordingly.

According to nature of variable, there may be three types of frequency distribution, namely:

1. Frequency distribution for qualitative or categorical variable,
2. Frequency distribution for discrete variable and
3. Frequency distribution for continuous variable.

3.1. Frequency distribution of categorical or qualitative data.

Frequency distribution for categorical variable is a table where observations of a qualitative variable are divided into different mutually exclusive categories. Here each category is called class. Usually it contains three columns. Different categories of the qualitative variable as class are in the first column, observations fall in each class presented by tally marks are in the second column, and frequencies for each class are shown in the third column.

Example 1.0. The following data represent the sizes of shirts sold by a departmental store during one week:

$$M, S, L, XL, M, M, M, S, S, S, M, M, XL, XL, L, M, M, M, M, S, S, L, L, XL,$$

$$M, M, M, S, S, L, L, S, M, M, L, L, M, M, L, XL, S, S, S, M, M, M, S, XL, L, L$$

$$M, M, M, M, S, M, M, S, XL, L.$$

Here, S, M, L, and XL denote small, medium, large, and extra-large shirts, respectively. Construct a frequency distribution table and comment on the results.

Solution. The frequency distribution of the given data is presented in the following table.

Shirt Size	**Tally Marks**	**Frequency**
Small (S)	𝍸 𝍸 𝍸	15
Medium (M)	𝍸 𝍸 𝍸 𝍸 𝍸 |	26
Large (L)	𝍸 𝍸 ||	12
Extra Large (XL)	𝍸 ||	7
Total		**60**

Comment: Medium-sized shirts have the highest demand, with 26 shirts sold, while extra-large shirts have the lowest demand, with 7 shirts sold. Therefore, the store should maintain a larger stock of medium-sized shirts to meet customer demand.

Example 2.0: The administration of a factory wants to assess the health status of its workers. A survey was conducted, and the health conditions of the workers were recorded as follows:

G, P, A, P, P, A, A, A, A, A, A, A, A, A, A, P, G, G, P, A, A, P, G, P, A, P,

G, A, G, P, A, A, A, A, G, P, P, A, A, G, P, A, P, A, G, A, A, A, A, P, P, G.

Here, G, A, and P denote good, average, and poor health, respectively. Construct a frequency distribution table and comment on the results.

Solution: The frequency distribution of the workers' health conditions is presented in the following table.

Health Condition	**Tally Marks**	**Frequency**
Good (G)	𝍸 𝍸	10
Average (A)	𝍸 𝍸 𝍸𝍸 𝍸 |	27
Poor (P)	𝍸 𝍸 𝍸	15
Total		**52**

Comment: The table shows that most workers (27) have average health, while 10 workers have good health and 15 workers have poor health. Therefore, the factory administration should take appropriate measures to improve the workers' health conditions.

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3.2. Diagrams for categorical data

Once the qualitative data have been categorized and summarized in a frequency distribution table, the next step is to display the frequency distribution of the qualitative data by different diagrams. Important diagrams for presenting categorical data are:

(i) Bar diagram,

(ii) Pie-diagram,

(iii) Pareto diagram,

(iv) Pictogram.

3.2.1 Bar diagram.

There are three types of bar diagram. They are:

(i) Simple bar diagram,

(ii) Multiple bar diagram and

(iii) Component bar diagram.

Here we shall discuss the diagrams one by one.

4.3.2. Simple bar diagram.

Simple bar diagram is the most popular diagrammatical representation of qualitative data. By simple bar diagram, only one qualitative variable can be exhibited. First a frequency distribution table for qualitative data is constructed. Then a bar of fixed width above each category or class is drawn. Actually bars are looked like rectangles of equal width over classes. The height of the bar over each class is equal to the class frequency. The bars should be separated by uniform gap to emphasize the fact that each class (category) is separate. Sometimes, bar diagrams are drawn for relative frequency or percent frequency distributions too. In those cases, the heights of the bars are taken proportional to class relative or percent frequencies.

Bar diagrams are widely used because they are easy to construct and easy to understand. They are effective tools for visual interpretations or comparison of data. Television, newspapers, magazines, books and reports make substantial use of this diagram to visually communicate to the viewer.

The main limitation of these diagrams is that only one qualitative variable can be displayed through these. More than one qualitative variables can be exhibited by multiple or component bar diagrams.

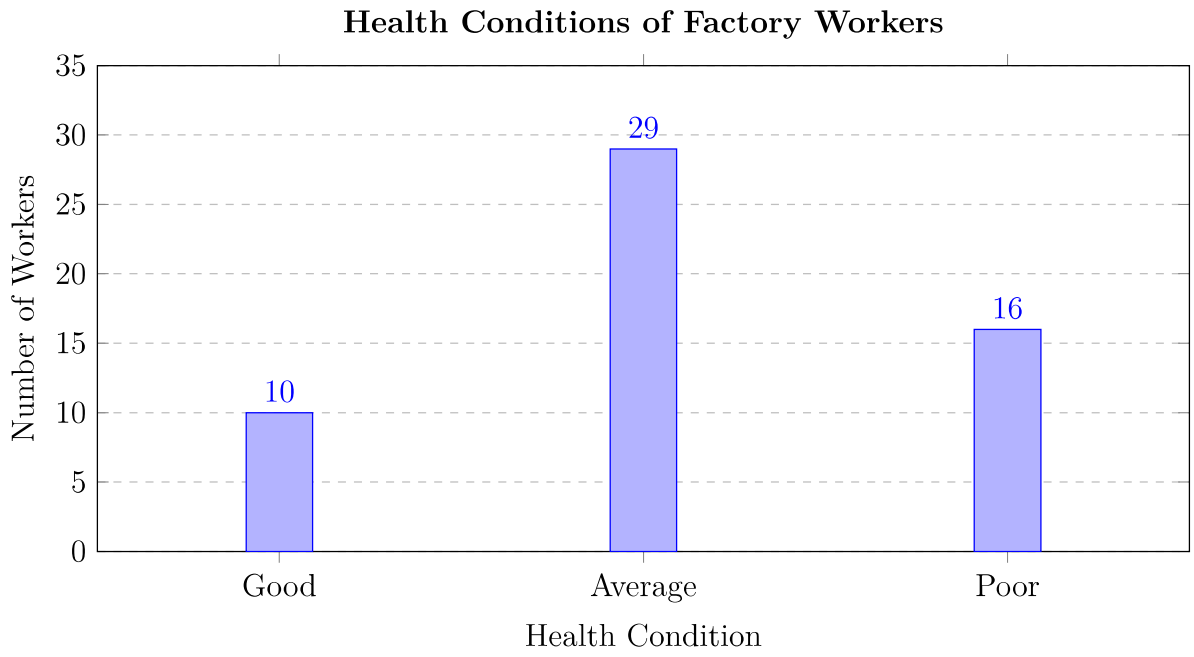
Example 03. The frequency distribution of the health condition of 55 workers of a factory is given below:

Category (Health Condition)	**Frequency (Number of Workers)**
Good	10
Average	29
Poor	16
Total	55

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Construct a bar diagram from the given frequency distribution.

Solution: To construct the bar diagram, we take the health conditions (Good, Average, and Poor) along the horizontal axis and the number of workers along the vertical axis. The heights of the bars are determined by the respective frequencies. The required bar diagram is shown below.



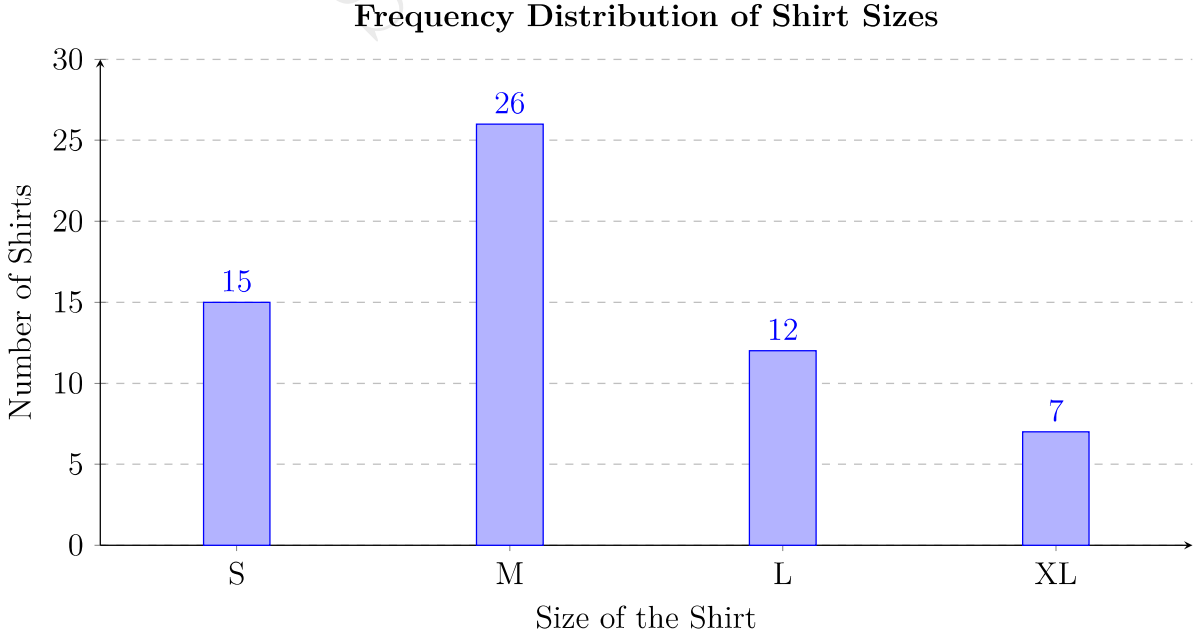
The bar diagram shows that 10 workers have good health, 29 have average health, and 16 have poor health. The largest number of workers belong to the average health category.

Example 04. Construct a bar diagram with the following frequency table.

Size of the shirt	S	M	L	XL
Frequency (number of shirts)	15	26	12	7

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Solution: To construct the bar diagram, we take the sizes of the shirts (S, M, L, and XL) along the horizontal axis and the number of shirts along the vertical axis. The heights of the bars are determined by the respective frequencies. The required bar diagram is shown below.



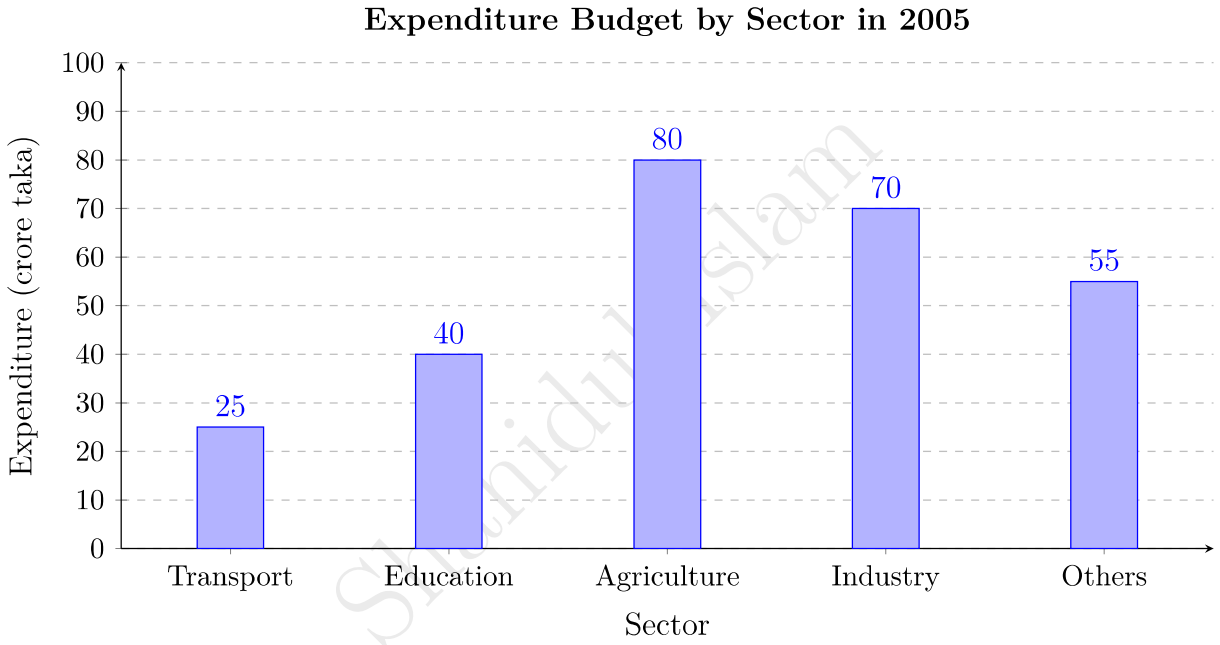
The bar diagram shows that the largest number of shirts belong to size M, while the smallest number belong to size XL.

Example 05. The following data give the expenditure budget in crore taka of different sectors of a country for the financial year 2005.

Sector	Transport	Education	Agriculture	Industry	Others	Total
Expenditure	25	40	80	70	55	270

Solution:

To construct the bar diagram, we take the different sectors (Transport, Education, Agriculture, Industry, and Others) along the horizontal axis and the expenditure in crore taka along the vertical axis. The heights of the bars are determined by the respective expenditures. The total expenditure is 270 crore taka and is not represented as a separate bar. The required bar diagram is shown below.



The bar diagram shows that the expenditure on Agriculture was the highest, amounting to 80 crore taka, followed by Industry at 70 crore taka and Others at 55 crore taka. Education received 40 crore taka, while Transport received the lowest allocation of 25 crore taka.

3.2.2 Multiple bar diagram

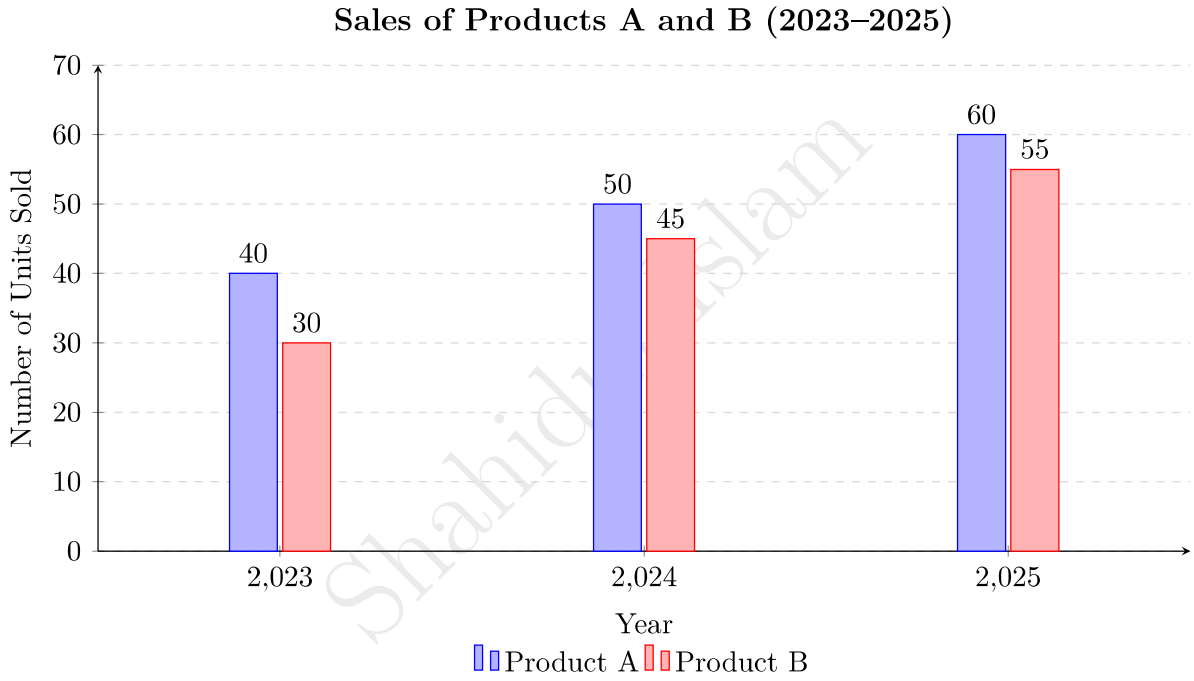
The multiple bar diagram is a graphical method of presenting two or more related sets of data using groups of bars drawn side by side. Each group represents a particular category, and each bar represents a different set of data. The bars are drawn with equal widths, and their heights are proportional to the corresponding values.

Example 06. The following table shows the sales of two products in a departmental store over three years:

Year	Product A	Product B
2023	40	30
2024	50	45
2025	60	55

Represent the data by a suitable multiple bar diagram.

Solution: To construct the multiple bar diagram, we take the years 2023, 2024, and 2025 along the horizontal axis and the number of units sold along the vertical axis. For each year, two adjacent bars represent the sales of Product A and Product B, respectively. Different colours are used to distinguish the two products. The heights of the bars are determined by their respective sales figures. The required multiple bar diagram is shown below.



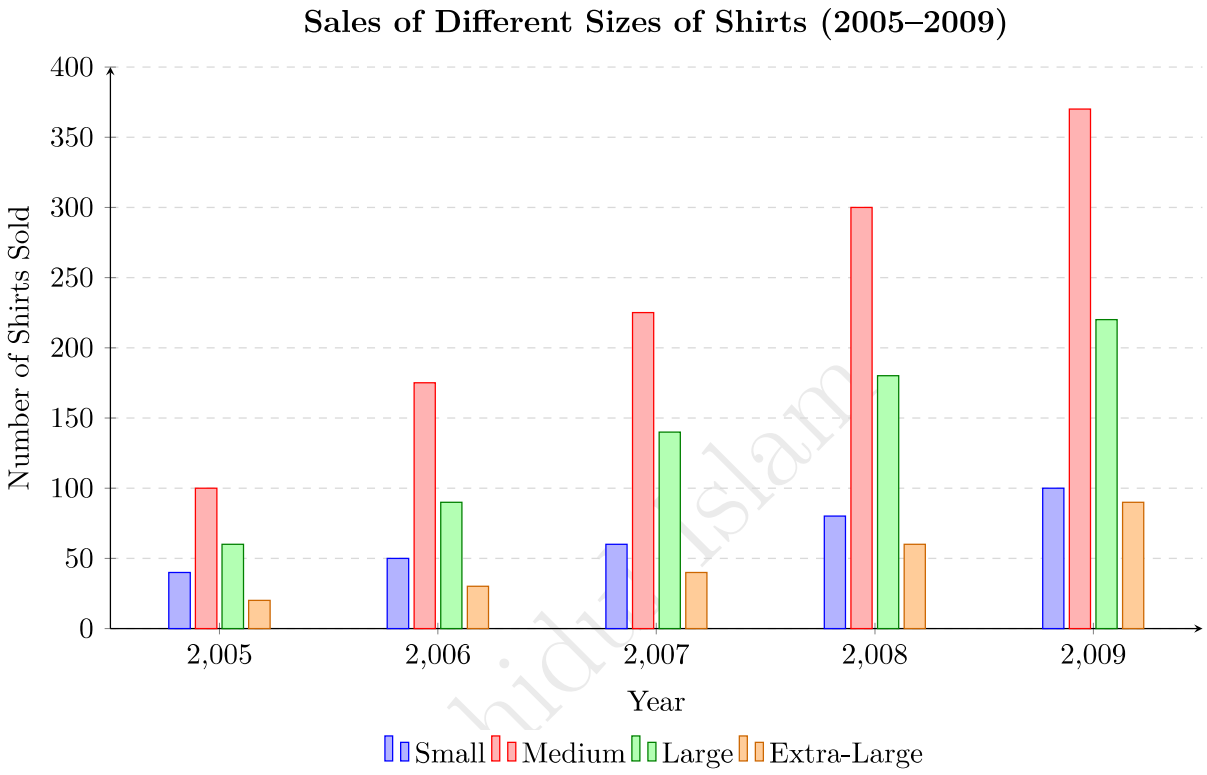
The multiple bar diagram shows that sales of both products increased over the three-year period. Product A had higher sales than Product B in each year. The sales of Product A increased from 40 units in 2023 to 60 units in 2025, while those of Product B increased from 30 units to 55 units over the same period.

Example 07. The following data give the pieces of different sizes of shirts sold by departmental stores for the last five years.

Year / Size of the shirt	2005	2006	2007	2008	2009
Small	40	50	60	80	100
Medium	100	175	225	300	370
Large	60	90	140	180	220
Extra-Large	20	30	40	60	90

Construct a multiple bar diagram to compare the sales of different sizes of shirts for the last five years.

Solution: To construct the multiple bar diagram, we take the years 2005, 2006, 2007, 2008, and 2009 along the horizontal axis and the number of shirts sold along the vertical axis. For each year, four adjacent bars represent the sales of Small, Medium, Large, and Extra-Large shirts, respectively. Different colours are used to distinguish the four sizes. The heights of the bars are determined by their respective frequencies. The required multiple bar diagram is shown below.



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The multiple bar diagram shows that sales of all four sizes of shirts increased over the five-year period. Medium shirts had the highest sales in every year, increasing from 100 pieces in 2005 to 370 pieces in 2009. Extra-Large shirts had the lowest sales in each year. Overall, the diagram indicates a steady growth in the sales of shirts of different sizes.

3.2.3 Component bar diagram

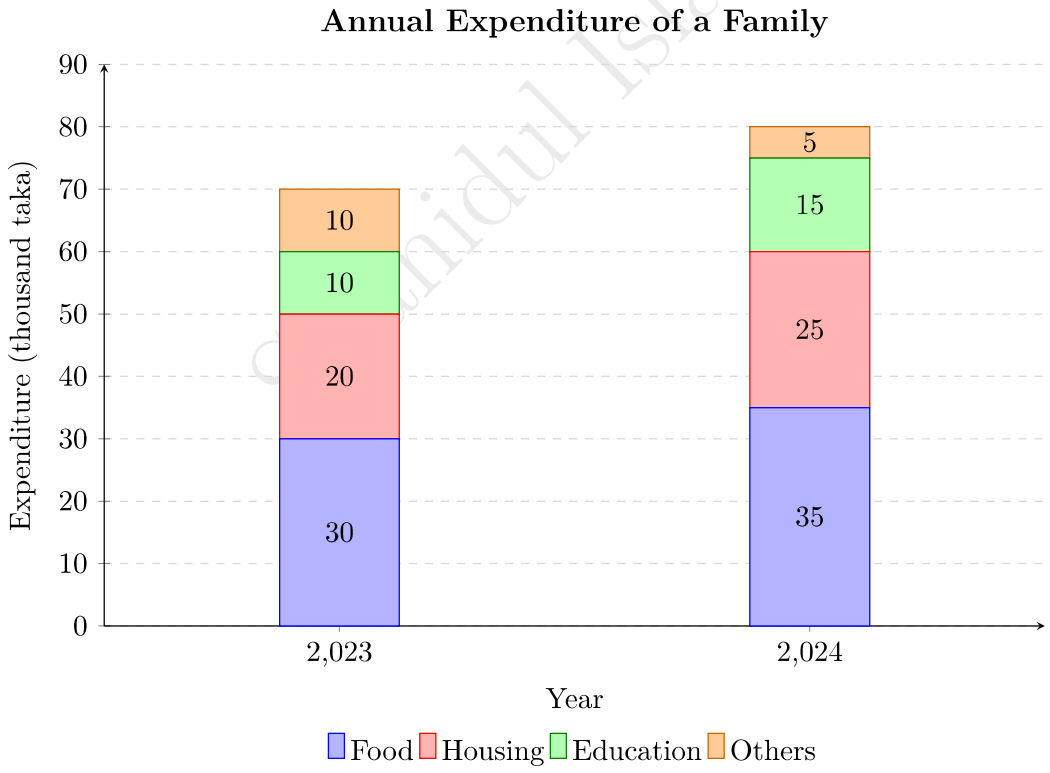
A component bar diagram is a graphical method of presenting data in which each bar represents the total value of a category and is divided into different segments (components). Each segment represents a part of the total, and the height or length of each segment is proportional to its corresponding value.

Example 08. The following table shows the annual expenditure of a family (in thousand taka):

Expenditure	2023	2024
Food	30	35
Housing	20	25
Education	10	15
Others	10	5
Total	70	80

Construct a suitable component bar diagram.

Solution: To construct the component bar diagram, we take the years 2023 and 2024 along the horizontal axis and the annual expenditure in thousand taka along the vertical axis. For each year, a single bar is divided into four components representing Food, Housing, Education, and Others. The height of each component is proportional to the corresponding expenditure, and the total height of each bar represents the total annual expenditure. Different colours are used to distinguish the components. The required component bar diagram is shown below.



The component bar diagram shows that the family's total annual expenditure increased from 70 thousand taka in 2023 to 80 thousand taka in 2024. Expenditure on Food, Housing, and Education increased, while expenditure on Others decreased from 10 thousand taka to 5 thousand taka. Food accounted for the largest share of expenditure in both years.

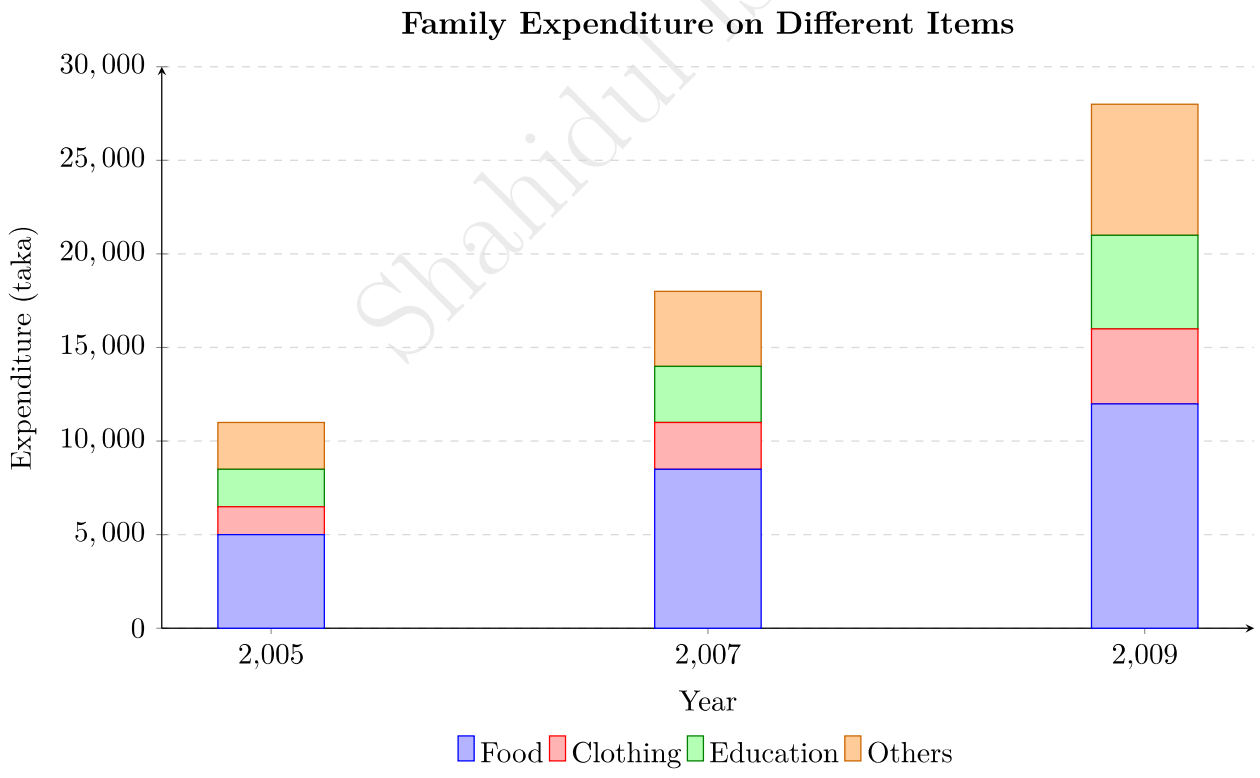
Example 09. The following table gives the expenditures in taka of a family on different items for the three years:

Item	Year	2005	2007	2009
	Food	5000	8500	12000
	Clothing	1500	2500	4000
	Education	2000	3000	5000
	Others	2500	4000	7000

Draw a component bar diagram with the above data.

Solution: To construct the component bar diagram, we take the years 2005, 2007, and 2009 along the horizontal axis and the expenditure in taka along the vertical axis. For each year, a single bar is divided into four components representing Food, Clothing, Education, and Others. The height of each component is proportional to the corresponding expenditure, and the total height of each bar represents the total expenditure for that year. Different colours are used to distinguish the components. The required component bar diagram is shown below.

The total expenditures are:
- 2005: $5000 + 1500 + 2000 + 2500 = 11000$ taka. - 2007: $8500 + 2500 + 3000 + 4000 = 18000$ taka.
- 2009: $12000 + 4000 + 5000 + 7000 = 28000$ taka.



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The component bar diagram shows that the family's total expenditure increased from 11,000 taka in 2005 to 18,000 taka in 2007 and 28,000 taka in 2009. Food accounted for the largest share of expenditure in all three years. Expenditure on every item increased over the period, indicating a steady rise in the family's overall expenditure.

3.2.4 Pie diagram or chart

The pie diagram or pie chart is a circular diagram divided into sectors, where each sector represents a component of the total and its size is proportional to the value of that component. The entire circle represents the total value, or 100%, and its central angle is 360°.

The central angle of each sector is calculated by

$$\text{Central Angle} = \frac{\text{Component Value}}{\text{Total Value}} \times 360°.$$

Example 10. The following table shows the annual expenditure of a family:

Expenditure	Amount (in thousand taka)
Food	30
Housing	20
Education	15
Clothing	10
Others	25
Total	100

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Represent the data by a suitable pie diagram.

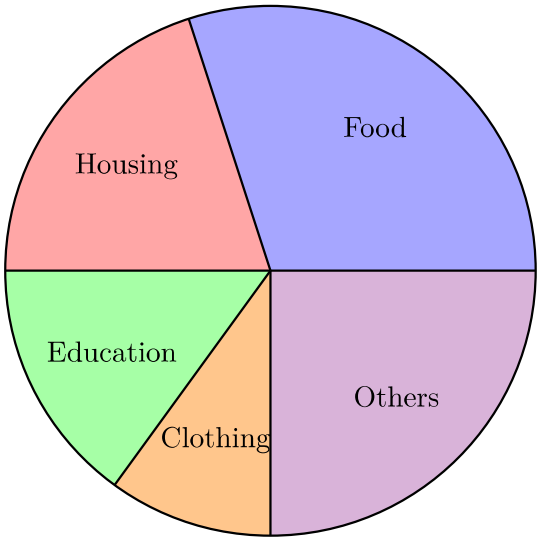
Solution: To construct the pie diagram, we first calculate the central angle for each item of expenditure. The central angle is obtained by multiplying the proportion of expenditure on each item by 360°.

$$\text{Central angle} = \frac{\text{Expenditure on an item}}{\text{Total expenditure}} \times 360°$$

The central angles are calculated as follows:

Expenditure	**Amount**	**Central Angle**
Food	30	$\frac{30}{100} \times 360° = 108°$
Housing	20	$\frac{20}{100} \times 360° = 72°$
Education	15	$\frac{15}{100} \times 360° = 54°$
Clothing	10	$\frac{10}{100} \times 360° = 36°$
Others	25	$\frac{25}{100} \times 360° = 90°$
Total	100	360°

The required pie diagram is shown below.



The pie diagram shows the distribution of the family's annual expenditure among five items. Food accounts for the largest share (30%), followed by Others (25%), Housing (20%), Education (15%), and Clothing (10%). The central angles of all sectors add up to 360°.

Example 11. The following data give the expenditure budget in crore taka of different sectors of a country for the financial year 2005.

Sector	Transport	Education	Agriculture	Industry	Others	Total
Expenditure	25	40	80	70	55	270

Construct a pie diagram for the given data.

Solution: To construct a pie diagram, we calculate the central angle corresponding to each sector using the formula

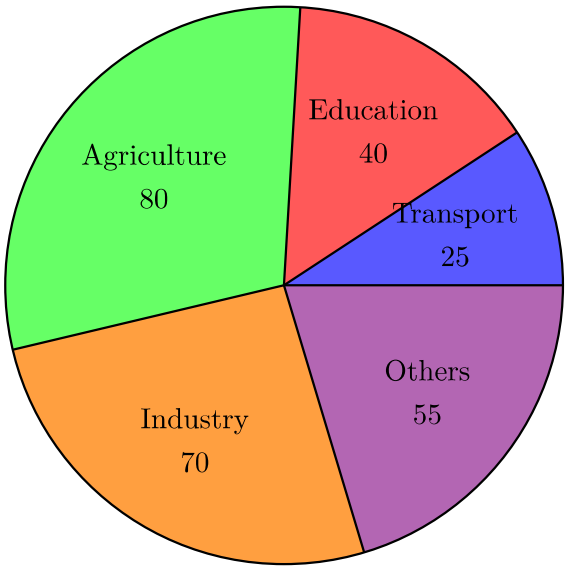
$$\text{Central angle} = \frac{\text{Sector expenditure}}{\text{Total expenditure}} \times 360^\circ.$$

Here, the total expenditure is 270 crore taka.

Sector	**Expenditure (crore taka)**	**Central Angle (degrees)**
Transport	25	$\frac{25}{270} \times 360^\circ = 33.33^\circ$
Education	40	$\frac{40}{270} \times 360^\circ = 53.33^\circ$
Agriculture	80	$\frac{80}{270} \times 360^\circ = 106.67^\circ$
Industry	70	$\frac{70}{270} \times 360^\circ = 93.33^\circ$
Others	55	$\frac{55}{270} \times 360^\circ = 73.34^\circ$
Total	**270**	**360°**

Using these central angles, the pie diagram can be constructed as follows.

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Agriculture accounts for the largest share of the country's expenditure budget, at 80 crore taka, while transport accounts for the smallest share, at 25 crore taka.

3.2.5 Pareto diagram

The Pareto diagram is a graphical representation of categorical data using vertical bars arranged in descending order of frequency or magnitude, from left to right. It helps identify the most significant categories contributing to a problem or situation.

Construction of a Pareto Diagram: The following steps are used to construct a Pareto diagram:

1. Collect data and classify them into different categories.
2. Calculate the frequency or magnitude of each category.
3. Arrange the categories in descending order of frequency or magnitude.
4. Draw two perpendicular axes. Represent the categories along the horizontal axis and their frequencies along the vertical axis.
5. Draw vertical bars of equal width, with heights proportional to the corresponding frequencies.
6. Label the axes and provide a suitable title for the diagram.

Example 12. Suppose in a University every department of Science has equal number of seats (say 50) for new enrollments. The following table gives the number of students applied for admission in different departments for the session 2007-2008.

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Name of the departments	**Number of students applied**
Mathematics	1225
Physics	1410
Chemistry	2418
Statistics	1560
Computer Science	2710
Botany	2352
Zoology	2360
Biochemistry	2548
Micro-biology	2570

Construct a Pareto diagram and comment.

Solution: A Pareto diagram is a graphical representation in which bars are arranged in descending order of frequency, accompanied by a line representing the cumulative percentage.
The total number of applicants is

$$\begin{aligned} N &= 1225 + 1410 + 2418 + 1560 + 2710 \\ &\quad + 2352 + 2360 + 2548 + 2570 \\ &= 19153. \end{aligned}$$

The cumulative percentage is calculated by

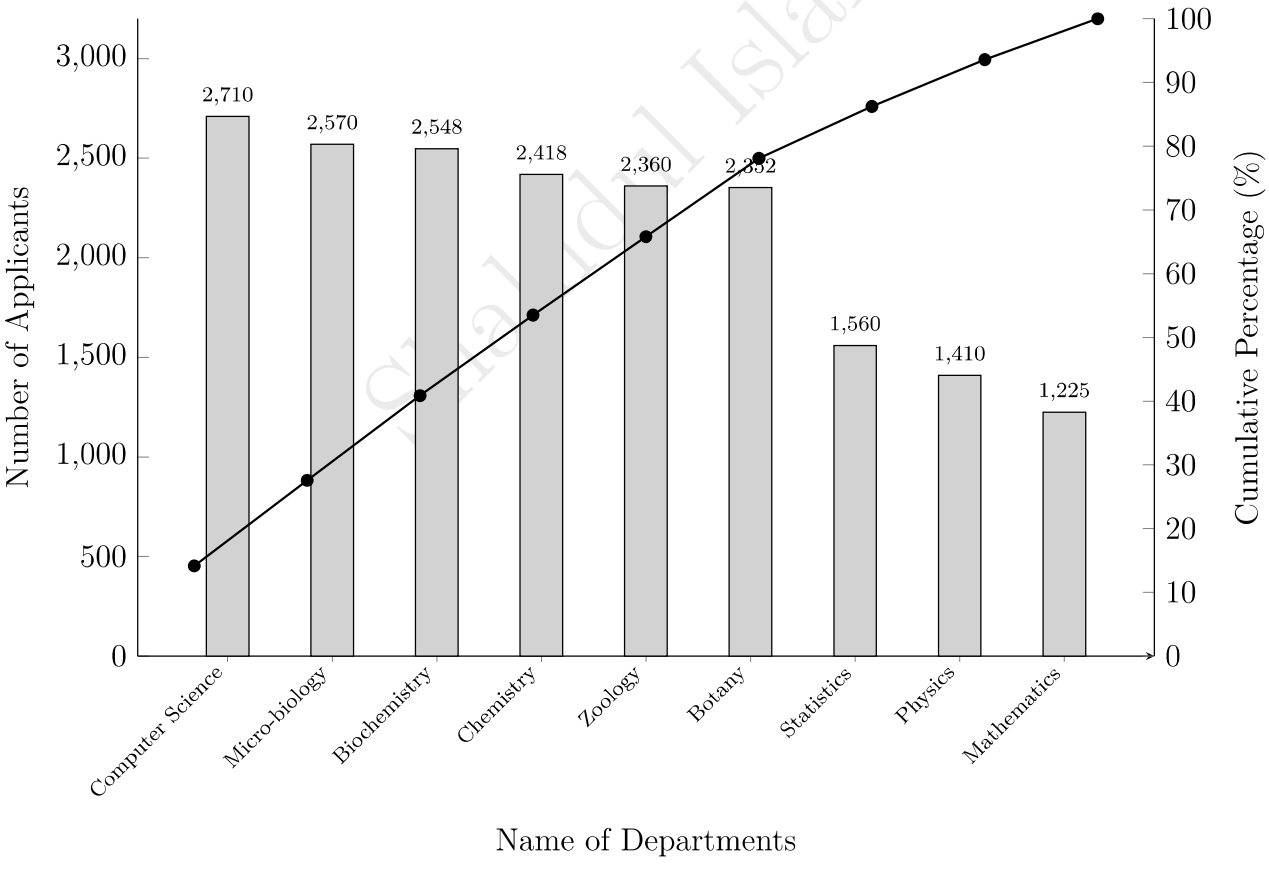
$$\text{Cumulative Percentage} = \frac{\text{Cumulative Frequency}}{\text{Total Frequency}} \times 100.$$

The required calculations are presented in the following table.

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Department	Applicants	Cumulative Frequency	Cumulative Percentage
Computer Science	2710	2710	14.15%
Micro-biology	2570	5280	27.57%
Biochemistry	2548	7828	40.87%
Chemistry	2418	10246	53.50%
Zoology	2360	12606	65.82%
Botany	2352	14958	78.10%
Statistics	1560	16518	86.24%
Physics	1410	17928	93.60%
Mathematics	1225	19153	100.00%
Total	**19153**		

The corresponding pareto diagram is constructed below.



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Comments:

1. Computer Science has the highest number of applicants, with 2710 students, whereas Mathematics has the lowest number, with 1225 students.

2. Computer Science, Micro-biology, and Biochemistry together account for approximately 40.87% of all applicants.

3. The first six departments, from Computer Science to Botany, account for approximately 78.10% of all applicants.

4. Each department has 50 seats. Therefore, Computer Science has the highest number of applicants per seat:
$$\frac{2710}{50} = 54.2.$$

5. Mathematics has the lowest number of applicants, but the number of applicants per seat is still
$$\frac{1225}{50} = 24.5.$$

The Pareto diagram shows that the first six departments account for approximately 78.10% of all applicants. Computer Science has the highest admission demand, with 2710 applicants for 50 available seats.

3.2.6 Pictogram

The pictogram is a graphical representation of statistical data using pictures or symbols, where each picture or symbol represents a fixed number of items. It provides a simple and attractive way to present and compare data.

Example 13. The following table refers to number of people of two countries for the last six censuses:

Year	1961	1971	1981	1991	2001	2011
Country A	3.5	5.5	6.5	7.0	7.5	8.0
Country B	2.0	4.0	5.5	7.0	8.0	8.5

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Draw a Pictogram with the data.

Solution: To represent the given population data, we choose the following key:

Key: ★ = 0.5 million people.

Thus, the number of symbols corresponding to each population value is obtained by dividing the population by 0.5. The required pictogram is presented below.

Year	Country A	Country B
1961	★★★★★★★	★★★★
1971	★★★★★★★★★★★	★★★★★★★★
1981	★★★★★★★★★★★★★	★★★★★★★★★★★
1991	★★★★★★★★★★★★★★	★★★★★★★★★★★★★★
2001	★★★★★★★★★★★★★★★	★★★★★★★★★★★★★★★
2011	★★★★★★★★★★★★★★★★	★★★★★★★★★★★★★★★★

Key: ★ = 0.5 million people.

Interpretation: The pictogram illustrates the population trends of the two countries over the six censuses. Country A had a larger population than Country B in 1961 and 1971. In 1991, the populations of the two countries were equal at 7.0 million. However, in 2001 and 2011, Country B had a larger population than Country A. Overall, the population of both countries increased throughout the period under consideration.

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3.3. Frequency distribution for quantitative data

The frequency distribution for quantitative data is a tabular arrangement of numerical observations that shows the number of times each value or group of values occurs in a dataset. It helps to organize, summarize, and interpret a large collection of numerical data systematically.

Quantitative data may be classified into two types:

1. **Discrete quantitative data:** Data that take distinct, countable values, such as the number of students in a class.
2. **Continuous quantitative data:** Data that can take any value within a given interval, such as height, weight, and temperature.

3.3.1 Ungrouped Frequency Distribution

An ungrouped frequency distribution presents each distinct numerical value along with its corresponding frequency.

Example: Consider the marks obtained by 10 students:

$$5,\ 6,\ 7,\ 5,\ 8,\ 6,\ 7,\ 5,\ 9,\ 6.$$

The frequency distribution is given below.

Marks (x)	**Frequency** (f)
5	3
6	3
7	2
8	1
9	1
Total	**10**

Here, the frequency represents the number of times each mark occurs, and the sum of all frequencies is equal to the total number of observations:

$$\sum_{i=1}^{k} f_i = n = 10.$$

3.3.2 Grouped Frequency Distribution

A grouped frequency distribution organizes numerical observations into a number of class intervals and records the frequency corresponding to each interval. It is particularly useful when the dataset contains many distinct values.

Example: The marks of 20 students are given below:

$$12,\ 15,\ 18,\ 21,\ 24,\ 27,\ 29,\ 31,\ 33,\ 35,$$
$$37,\ 39,\ 41,\ 43,\ 45,\ 47,\ 49,\ 52,\ 55,\ 58.$$

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Taking the class width as 10, we obtain the following grouped frequency distribution.

Class Interval	**Frequency** (f)
10–19	3
20–29	3
30–39	4
40–49	5
50–59	3
Total	**18**

The class intervals represent groups of numerical values, while their frequencies indicate the numbers of observations falling within the respective intervals.

Note

1. **Class interval:** A range used to group numerical observations, such as 10–19.
2. **Class limits:** The smallest and largest stated values of a class interval.
3. **Class width:** The difference between the upper and lower class boundaries.
4. **Class midpoint:** The arithmetic mean of the lower and upper class boundaries. It is given by
$$m_i = \frac{L_i + U_i}{2},$$
where L_i and U_i denote the lower and upper class boundaries, respectively.
5. **Frequency:** The number of observations belonging to a particular value or class interval.
6. **Relative frequency:** The ratio of the class frequency to the total number of observations, given by
$$r_i = \frac{f_i}{n}.$$
7. **Cumulative frequency:** The running total of frequencies up to a specified class, given by
$$F_i = \sum_{j=1}^{i} f_j.$$

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3.3.3 General Form of a Grouped Frequency Distribution

The general form of a grouped frequency distribution is presented in the following table.

Class Interval	Frequency (f_i)	Relative Frequency (f_i/n)	Cumulative Frequency (F_i)
a_1–b_1	f_1	f_1/n	f_1
a_2–b_2	f_2	f_2/n	$f_1 + f_2$
⋮	⋮	⋮	⋮
a_k–b_k	f_k	f_k/n	n
Total	$\boldsymbol{n}$	**1**	$\boldsymbol{n}$

A frequency distribution is useful for constructing histograms, frequency polygons, and ogives, as well as for calculating statistical measures such as the mean, median, mode, and standard deviation.

3.3.4 Procedure for the construction of a frequency distribution for quantitative Data

The following steps are generally followed to construct a frequency distribution for quantitative data.

1. **Collect the data:** Obtain the numerical observations from the given dataset.

2. **Determine the range:** Find the difference between the largest and smallest observations. Thus,
$$R = X_{\max} - X_{\min}.$$

3. **Determine the number of classes:** Decide the appropriate number of class intervals. Sturges' formula may be used as a guideline:
$$k = 1 + 3.322 \log_{10} n,$$
where k denotes the approximate number of classes and n is the total number of observations.

4. **Calculate the class width:** Determine the approximate width of each class by dividing the range by the number of classes:
$$h = \frac{R}{k}.$$
The class width may be rounded to a convenient value.

5. **Construct the class intervals:** Form suitable, mutually exclusive, and collectively exhaustive class intervals that cover all observations. The intervals should preferably have equal widths.

6. **Tally the observations:** Examine each observation and place a tally mark in the appropriate class interval.

7. **Determine the class frequencies:** Count the tally marks in each class to obtain its frequency, denoted by f_i.

8. **Verify the distribution:** Add all class frequencies and check that their sum equals the total number of observations:
$$\sum_{i=1}^{k} f_i = n.$$

9. **Calculate additional frequencies (if required):** Determine the relative frequencies and cumulative frequencies using
$$r_i = \frac{f_i}{n}, \qquad F_i = \sum_{j=1}^{i} f_j.$$

Following these steps produces a systematic and informative summary of quantitative data, facilitating statistical analysis and graphical representation.

Example 14. In a winter season, the temperature (in celsius) of the month of January in Srimangal is placed below. Find the frequency distribution table of the temperature.

$$14^\circ, 14^\circ, 14^\circ, 13^\circ, 12^\circ, 13^\circ, 10^\circ, 10^\circ, 11^\circ, 12^\circ, 11^\circ, 10^\circ, 9^\circ, 8^\circ, 9^\circ, 11^\circ, 10^\circ, 10^\circ,$$

$$8^\circ, 9^\circ, 7^\circ, 6^\circ, 6^\circ, 6^\circ, 6^\circ, 7^\circ, 8^\circ, 9^\circ, 9^\circ, 8^\circ, 7^\circ$$

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Solution: Here the minimum and maximum numerical values of the data of temperature are 6 and 14 respectively.
Hence the range $= (14 - 6) + 1 = 9$
If the class interval is considered to be 3, the numbers of class will be $\frac{9}{3} = 3$.
Considering 3 to be the class interval, if the data are arranged in 3 classes, the frequency table will be:

Temperature (in celcius)	Tally	Frequency
$6° - 8°$	𝍸 𝍸 |	11
$9° - 11°$	𝍸 𝍸 |||	13
$12° - 14°$	𝍸 ||	7
	Total	31

Example 15. The marks obtained in Mathematics by 40 students in an annual examination are given below. Make a cumulative frequency table of the marks obtained.

70, 40, 35, 60, 55, 58, 45, 60, 65, 80, 70, 46, 50, 60, 65, 70, 58, 60, 48, 70, 36, 85,

60, 50, 46, 65, 55, 61, 72, 85, 90, 68, 65, 50, 40, 56, 60, 65, 46, 76

Solution:

Range of the data = (highest numerical value - lowest numerical value) + 1 = (90 - 35) + 1 = 55 + 1 = 56

Let the class interval be 5 then the number of classes $= \frac{56}{5} = 11.2$ or 12 [taking the immediate next integer when class interval is fractional].

Hence the cumulative frequency distribution table at a class interval of 5 will be as follows:

Shahidul Islam

Obtained marks	Tally	Frequency	Cumulative frequency
35-39	||	2	$2+2=4$
40-44	||	2	$2+2=4$
45-49	𝍸	5	$5+4=9$
50-54	|||	3	$3+9=12$
55-59	𝍸	5	$5+12=17$
60-64	𝍸 ||	7	$7+17=24$
65-69	𝍸 |	6	$6+24=30$
70-74	𝍸	5	$5+30=35$
75-79	|	1	$1+35=36$
80-84	|	1	$1+36=37$
85-89	||	2	$2+37=39$
90-94	|	1	$1+39=40$

3.4. Graphical representation of quantitative data

The graphical representation of quantitative data is a method of presenting numerical information in a visual form to facilitate its understanding, interpretation, and analysis. It provides a clear and concise summary of the characteristics of a dataset and helps identify patterns, trends, distributions, and variations in the observations.

The important graphs for representing frequency distribution of quantitative data are:

1. Dot plot
2. Histogram
3. Frequency polygon
4. Frequency curve
5. Ogive polygon
6. Ogive curve
7. Stem-and-leaf plot
8. Line graph of time series data and
9. Scatter diagram

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3.5. Graphical Representation of Frequency Distribution of Quantitative Data

The graphical representation of quantitative data is an important statistical technique used to summarize, organize, and interpret numerical observations. A suitable graph presents the distribution of data in a visual form, enabling the reader to identify patterns, compare frequencies, examine variability, and understand relationships among variables.

The choice of a graphical method depends on the nature of the data, the purpose of the analysis, and the type of information to be communicated. The important graphical methods for representing quantitative data are discussed below.

3.5.1 Dot Plot

A **dot plot** is a graphical method in which each observation is represented by a dot placed above its corresponding value on a number line. When a value occurs more than once, the dots are stacked vertically to indicate its frequency.

A dot plot is particularly useful for small or moderately sized data sets because it displays individual observations while providing a clear picture of the distribution.

Example. Consider the following observations representing the number of books read by seven students in a month:

$$2,\ 3,\ 3,\ 4,\ 4,\ 4,\ 5.$$

The corresponding dot plot is shown in Figure 1.

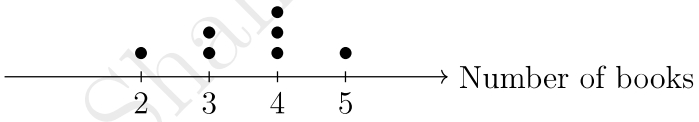


Figure 1: Dot plot of the number of books read by students.

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The value 4 has the highest frequency, occurring three times. The observations range from 2 to 5.

Applications: A dot plot is useful for identifying repeated values, clusters, gaps, the range of observations, and possible outliers.

3.5.2 Histogram

A **histogram** is a graphical representation of a frequency distribution for quantitative data, particularly continuous data. It consists of adjacent rectangles, where each rectangle represents a class interval and its height reflects the corresponding frequency when the class widths are equal.

Unlike a bar chart, a histogram has no gaps between adjacent rectangles because the class intervals represent a continuous numerical scale.

Example. Suppose the examination marks of 20 students are grouped as follows.

Marks	Frequency	Cumulative Frequency
0–10	2	2
10–20	4	6
20–30	7	13
30–40	5	18
40–50	2	20
Total	20	

The histogram is constructed by placing the class intervals on the horizontal axis and the corresponding frequencies on the vertical axis. Five adjacent rectangles are drawn, with heights $2, 4, 7, 5$, and 2, respectively.

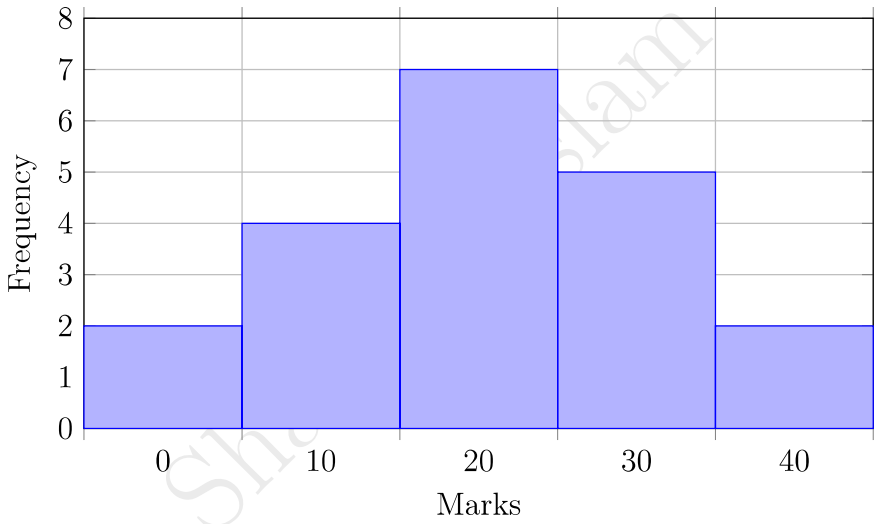


Figure 2: Histogram of examination marks.

The tallest rectangle corresponds to the interval 20–30, indicating that the largest number of students obtained marks in that interval.

When class widths are unequal, the height of each rectangle should represent the **frequency density**, defined by

$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Class width}}.$$

In this case, the area of each rectangle, rather than its height alone, is proportional to the frequency.

Applications: A histogram helps identify the shape of a distribution, its concentration, spread, symmetry, skewness, and possible multiple peaks.

3.5.3 Frequency Polygon

A **frequency polygon** is a line graph obtained by plotting the frequencies against the midpoints of the corresponding class intervals and joining the plotted points by straight-line segments. The two endpoints are generally connected to the horizontal axis at the midpoints of imaginary classes having zero frequencies, immediately before the first class and after the last class.

The midpoint of a class interval is calculated as

$$\text{Class midpoint} = \frac{\text{Lower class limit} + \text{Upper class limit}}{2}.$$

Example. Consider the frequency distribution of examination marks.

Table 1: Class midpoints and frequencies.

Class interval	Midpoint	Frequency
0–10	5	2
10–20	15	4
20–30	25	7
30–40	35	5
40–50	45	2

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The points to be plotted are

$$(5, 2),\ (15, 4),\ (25, 7),\ (35, 5),\ (45, 2).$$

To close the polygon, the additional points $(-5, 0)$ and $(55, 0)$ may be included.

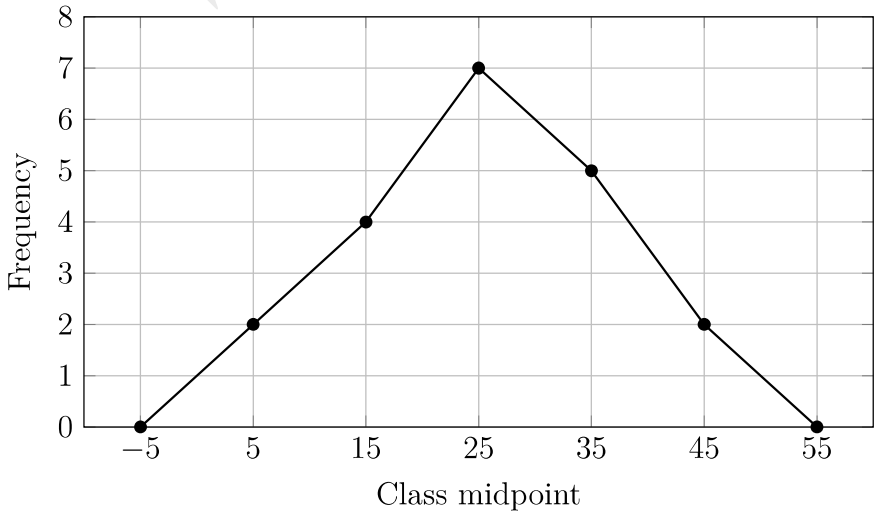


Figure 3: Frequency polygon of examination marks.

The polygon reaches its maximum height at the midpoint 25, where the frequency is 7. Thus, the class interval 20–30 contains the largest number of observations.

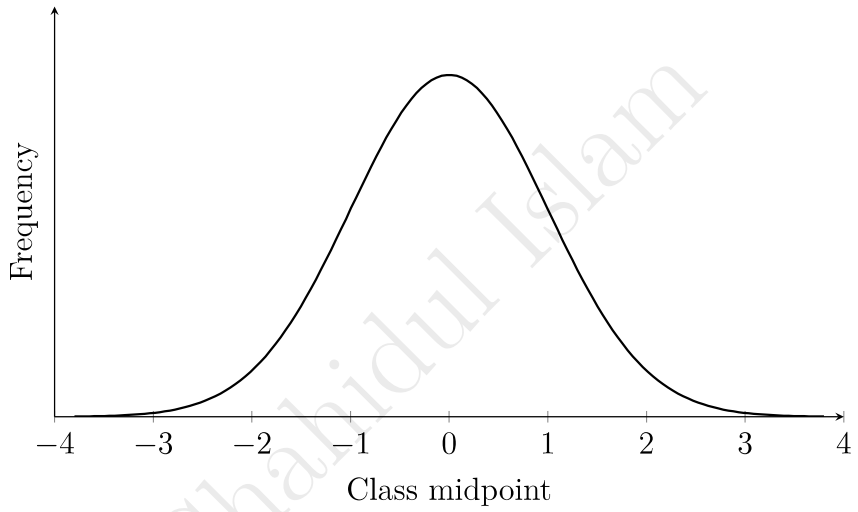
Applications: A frequency polygon is useful for examining the general shape of a distribution and comparing two or more frequency distributions on the same coordinate system.

3.5.4 Frequency Curve

A **frequency curve** is a smooth curve representing a frequency distribution. It is obtained by plotting class midpoints against their corresponding frequencies and drawing a smooth curve through or near the plotted points. The curve represents the overall pattern of the distribution rather than the individual fluctuations between adjacent class frequencies.

A frequency curve differs from a frequency polygon in that the latter consists of straight-line segments, whereas the former is represented by a smooth line.

Example. Consider a frequency distribution in which frequencies increase gradually toward the middle class and then decrease toward the higher classes. A frequency curve may take the form of a bell-shaped curve when the distribution is approximately symmetric and unimodal.



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Figure 4: Illustrative bell-shaped frequency curve.

The curve rises as the frequencies increase, reaches a maximum near the centre, and then declines as the frequencies decrease. This indicates that observations are concentrated around the central values.

A frequency curve need not be bell-shaped. Depending on the data, it may be positively skewed, negatively skewed, J-shaped, or multimodal.

Applications: Frequency curves are useful for describing the general form of a distribution, studying its symmetry or skewness, and comparing the overall patterns of different distributions.

3.5.5 Ogive Polygon

An **ogive polygon**, commonly called a cumulative frequency polygon, is a graphical representation of cumulative frequencies. It is constructed by plotting cumulative frequencies against the appropriate class boundaries and joining successive points by straight-line segments.

There are two principal types of cumulative frequency distributions:

1. Less-than cumulative frequency distribution.

2. More-than cumulative frequency distribution.

A less-than ogive polygon uses the upper class boundaries and the corresponding less-than cumulative frequencies. A more-than ogive polygon uses the lower class boundaries and the corresponding more-than cumulative frequencies.

Example. Consider the following distribution of examination marks.

Marks	Frequency	Less-than cumulative frequency
0–10	2	2
10–20	4	6
20–30	7	13
30–40	5	18
40–50	2	20
Total	20	

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For a less-than ogive polygon, plot the points

$$(0,0),\ (10,2),\ (20,6),\ (30,13),\ (40,18),\ (50,20).$$

The points are then joined by straight-line segments.

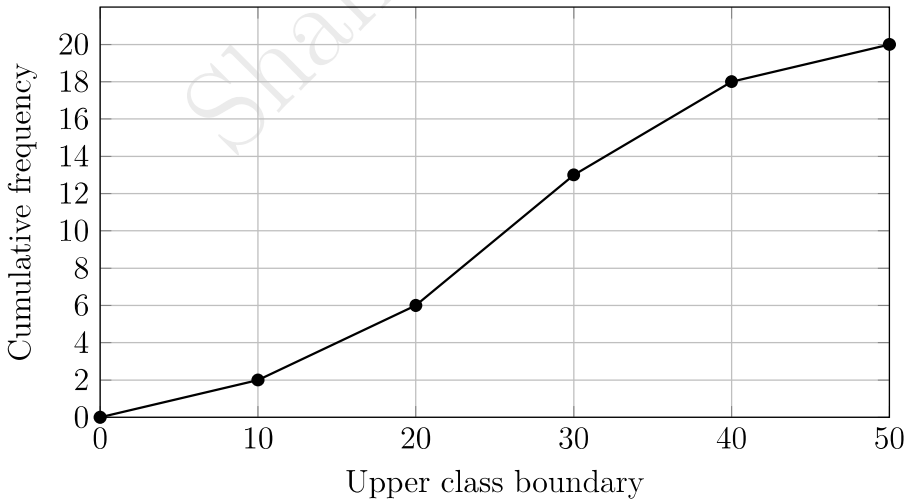


Figure 5: Less-than ogive polygon.

The cumulative frequency at the upper boundary 30 is 13. Therefore, 13 students obtained marks below 30, under the stated class-interval convention.

Applications: An ogive polygon is useful for determining cumulative frequencies, locating the median and other positional measures, and estimating the number or proportion of observations below or above a specified value.

3.5.6 Ogive Curve

An **ogive curve** is a smooth graphical representation of a cumulative frequency distribution. It is obtained by plotting cumulative frequencies against the corresponding class boundaries and drawing a smooth curve to represent the overall cumulative pattern.

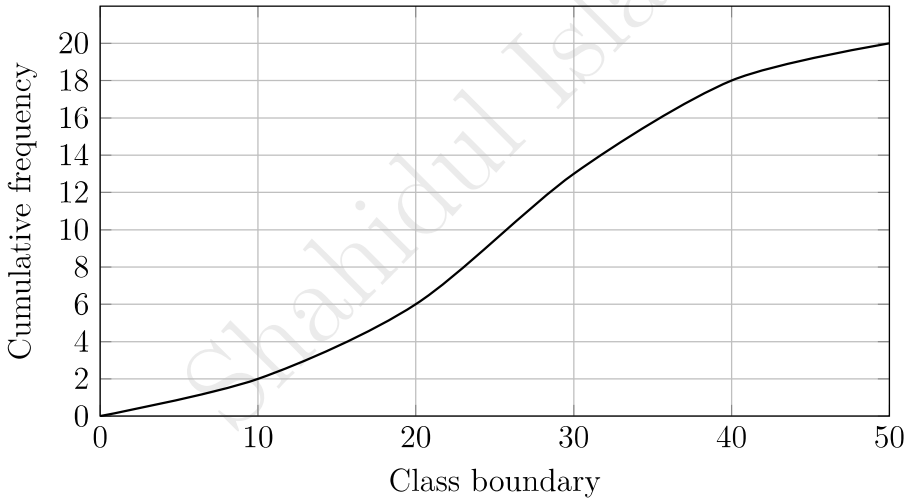
Like an ogive polygon, an ogive curve may be classified into two types:

- **Less-than ogive curve:** Represents the cumulative number of observations below the specified class boundaries. It generally increases from left to right.
- **More-than ogive curve:** Represents the cumulative number of observations above the specified class boundaries. It generally decreases from left to right.

Example. Consider the cumulative frequency distribution of examination marks presented in the previous section. A less-than ogive curve can be drawn by plotting the points

$$(0, 0),\ (10, 2),\ (20, 6),\ (30, 13),\ (40, 18),\ (50, 20)$$

and joining them with a smooth increasing curve.



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Figure 6: Illustrative less-than ogive curve.

The distinction between an ogive polygon and an ogive curve is primarily graphical. The former connects the plotted points with straight-line segments, whereas the latter uses a smooth curve to represent the cumulative distribution.

An important application of ogives is the determination of the median. For a distribution containing N observations, the median corresponds approximately to the cumulative frequency $N/2$. In the present example, $N = 20$, so the median is associated with cumulative frequency 10. By locating 10 on the vertical axis and projecting to the ogive, the corresponding mark can be estimated.

Applications: Ogive curves are useful for estimating the median, quartiles, percentiles, and the proportion of observations lying below or above a specified value.

3.5.7 Stem-and-Leaf Plot

A **stem-and-leaf plot** is a graphical technique for displaying quantitative data by dividing each observation into two parts: a stem and a leaf. The stem generally consists of the leading digit or digits, while the leaf consists of the final digit.

This method preserves the original observations while arranging them systematically, making it particularly suitable for small data sets.

Example. Consider the following examination marks obtained by ten students:

$$12,\ 15,\ 17,\ 21,\ 23,\ 24,\ 26,\ 31,\ 34,\ 38.$$

The corresponding stem-and-leaf plot is given below.

Stem	Leaves
1	2 5 7
2	1 3 4 6
3	1 4 8

Table 2: Stem-and-leaf display of examination marks.
Key: $1 \mid 2 = 12$ *marks.*

In this display, the stem 1 represents marks in the teens, and its leaves $2, 5, 7$ represent $12, 15, 17$, respectively. Similarly, the stem 2 represents marks in the twenties, and the stem 3 represents marks in the thirties.

The observations can be reconstructed directly from the plot. Furthermore, the concentration and spread of the marks can be examined without losing the individual values.

Applications: A stem-and-leaf plot is useful for organizing raw data, identifying the centre and spread of a distribution, detecting gaps and potential outliers, and preserving individual observations for further analysis.

3.5.8 Line Graph of Time Series Data

A **line graph of time series data** is a graphical representation used to show how a quantitative variable changes over time. Time is placed on the horizontal axis, while the observed values of the variable are placed on the vertical axis. The successive observations are joined by straight-line segments.

A time series consists of observations recorded at successive time points, such as daily temperatures, monthly rainfall, annual production, or yearly population.

Example. Suppose the annual production of a factory, measured in thousands of units, is recorded as follows.

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Table 3: Annual factory production.

Year	Production (thousand units)
2021	20
2022	25
2023	23
2024	30
2025	35

The points $(2021, 20)$, $(2022, 25)$, $(2023, 23)$, $(2024, 30)$, $(2025, 35)$ are plotted on the coordinate system and joined successively.

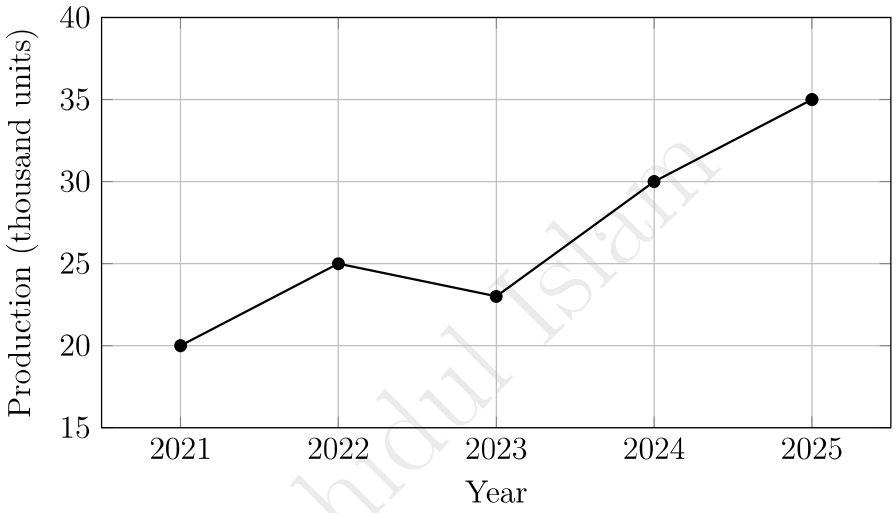


Figure 7: Annual factory production from 2021 to 2025.

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The graph indicates an overall upward trend in production over the five-year period, although production declined slightly from 2022 to 2023. Production increased from 20,000 units in 2021 to 35,000 units in 2025.

Applications: A time series line graph is useful for identifying trends, seasonal variations, fluctuations, turning points, and changes in a variable over time. It is also widely used for forecasting and monitoring economic, industrial, environmental, and demographic indicators.

3.5.9 Scatter Diagram

A **scatter diagram**, also known as a scatter plot, is a graphical method for examining the relationship between two quantitative variables. Each observation is represented by a point whose horizontal and vertical coordinates correspond to the values of the two variables.

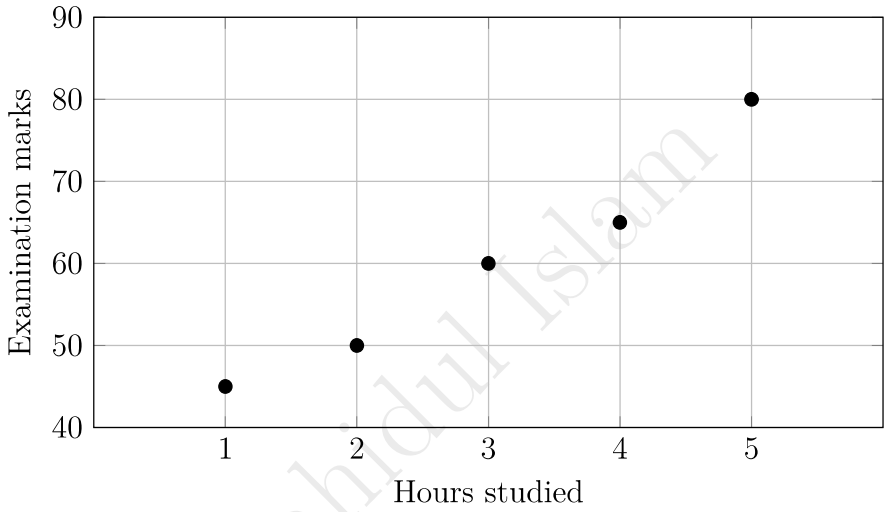
A scatter diagram helps determine whether the variables exhibit a positive association, a negative association, or little apparent association.

Example. Consider the following hypothetical data showing the hours studied and examination marks obtained by five students.

Table 4: Study hours and examination marks.

Student	Hours studied (x)	Marks (y)
A	1	45
B	2	50
C	3	60
D	4	65
E	5	80

The ordered pairs are $(1, 45),\ (2, 50),\ (3, 60),\ (4, 65),\ (5, 80)$. Plotting these points produces the scatter diagram shown in Figure 8.



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Figure 8: Relationship between study hours and examination marks.

The points generally rise from left to right, indicating a **positive association** between study hours and examination marks in this example. In general, a positive association means that larger values of one variable tend to occur with larger values of the other. A negative association occurs when larger values of one variable tend to accompany smaller values of the other.

A scatter diagram may also reveal a nonlinear relationship, clusters, or unusual observations. However, an apparent association does not, by itself, establish a causal relationship between the variables.

Applications: Scatter diagrams are useful for preliminary correlation analysis, identifying relationships between variables, detecting unusual observations, and assessing whether regression analysis may be appropriate.